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Aditi Sarkar

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With 7+ years of experience in both the 4-wheeler and 2-wheeler domains, I've specialized in cutting-edge technologies such as Advanced Driver Assistance Systems (ADAS), e-Powertrain, High Voltage Battery Management Systems (BMS), Charging and VCU. I have played a key role in developing HiL architectures, building tools that simplify everyday workflows and designing automated test scenarios for complex system-level validations.

Work experience

Senior Automation Engineer — **Ultraviolette Automotive** — **July 2025 – Present**

Currently working in the Embedded Team

- Designed and deployed an automated **HIL validation framework** using Robot Framework and Python for HV BMS, eliminating Vector CANoe license dependency and reducing manual validation effort by **95%**.
- Authored **200+** HV BMS test cases covering cell balancing, protection logic, and safety-critical requirements. This helped in identifying critical **20+** functional defects at pre-integration.
- Developed **Python-based simulator APIs** using hardware schematics and datasheets to validate different BMS scenarios. Automated programmable power supply control through Python scripting, enabling **100%** automated execution of HIL Test Framework.
- Built a PySide6-based system-level test UI and a Test Runner tool, reduced debugging turnaround time by **80%** and supported HV BMS HIL Test Jig stabilization and **cross-team issue resolution**.
- **Led** and strengthened **HV BMS EOL software** by refining requirements, restructuring test logic, enhancing backend execution, and adding flashing and cell calibration capability, enabling **100+** HV BMS and CMB board validations per day and improving First-Pass Yield from **80% to 97%**.
- Developed an HV BMS Pack Configurator tool after researching **low-cost CAN interfaces**, automated barcode scanning, and integrating **AWS** (authentication and S3 bucket). Eliminated manual configuration errors in the factory across **10+ bike models**.
- Upgraded the DCP Test Suite using Python-CAN and Robot Framework for new DCP Firmware and **root-caused** a critical DCP current-limit issue in the latest firmware code.

Diagnostics & Automation — **Mercedes-Benz R&D** — **December'22 – June'25**

Worked with the V&V Team on IDC6, an advanced L2+ and L2++ ADAS technology

- Analyzed **630+** requirements and developed **350+** test cases (CAPL, OTX), improving test coverage by **5x**.
- Conducted ECU/HIL testing for **25+** releases, raising **30+** critical bug reports with a high acceptance rate.
- Independently **led** CyberHIL and Camera Calibration for Driving and Parking, including HIL setup and test sequencing.
- Led the **CDD Authoring** team, managing **DID's, RID's, DTC** updates based on the UDS services for IDC(Gen6) ECU Base Software.
- Developed **front-end GUI** and critical **backend components** supporting **10+** operations for SCN and parallel IDC6 teams.
- Implemented Python scripts for artifact comparison and validation using Pandas and NumPy, achieving a **12x** performance boost with zero-defect delivery.
- Contributed to **CI/CD** pipelines for concurrent automated file generation and CDD artifact verification.

Senior Software Engineer — **KPIT Technologies Ltd** — **November'19 – December'22**

Worked with MicroFuzzy-KPIT for client BMW in CCU Verification and Validation Team.

- Authored **100+** test cases and implemented CAPL scripts in CANoe for system validation across **6** carlines.
- Automated test execution, reducing manual effort by **60%** and **optimizing flashing process** with Trace32, E-Sys, and INCA through Python Scripts.

- Handled Smoke, Functional, and Regression testing across 6 major topics, including **Diagnostics, DTCs, NVM, Power Management, Security and Flashing**.
- **Managed** 4+ parallel releases with on-time delivery, zero-defect artifacts, and a CSAT score of **4.92**.

Education

Tamil Nadu, IN **VIT, Vellore** 🏛

- B.Tech in Electrical and Electronics Engineering, July 2019. CPI: **8.6** (on scale of 10).

Kolkata, IN **Kendriya Vidyalaya** 🏛

- XII standard (CBSE Board) in PCM + Computer Science. Grade: **92%**
- X standard (CBSE Board). CPI:**10** (on scale of 10).

Skills

Programming:
- Python, CAPL, Embedded C, C

Communication protocols:
- CAN, CANFD, Ethernet DoIP, UART, I2C, SPI

Tools:
- CANdela Studio, CANoe, Monaco DTS9, OTX Studio, Lauterbach,
- vTESTstudio, MATLAB, Keil, GIT, SVN, STARC, DOORS, JIRA

Python Frameworks:
- Pandas, Numpy, PyVisa, PySerial, Selenium, PyQt5, PySide6, Tkinter, Python-can, boto3, win32com
- **Testing Frameworks**:
  - Robot Framework, PyTest

Others:
- UDS (ISO 14229), OBD (SAE J1979), FuSa, VT Systems, HIL Testing
- Variant Coding, Flashing Automation, CDD Authoring, ODX, Jenkins
- AWS, CI/CD Pipeline, Black Box Testing, Traceability

Achievements and Certifications

- **ISTQB®** - International Software Testing Qualifications Board Certified (Issued Sep, 2021)
- **High Flyers and KROWN Awards (5× Recipient)** - KPIT Technologies
- **Process Innovation Award** - Mercedes-Benz Research and Development India, for automation of concurrent CDD artifact generation and validation.
- **UV Spark Award** - Ultraviolette Automotive, for outstanding technical contributions within a span of a few months.
- **Course Certifications**:
  - Complete Python Pro
  - PyQt5
  - AUTOSAR Diagnostics (UDS, On-Board Diagnostics)
  - Adaptive AUTOSAR
  - CAN: Learn the Protocol

Interests

Technical blogging

- Write in-depth technical articles on Automotive Communication Protocols, Embedded Systems and Automation Frameworks.

Open-Source Projects

- Developed **CANableView** and **CANSerialGateway**, later adopted across 3 departments, delivering the same workflows as IXXAT/PCAN at about 0.05x the cost, with added features.